

1 Difficulty Screening

Five visual domains

Source pool

Chart

Table

GUI grounding

VisualSearch

Web-to-HTML

No-tool baseline
Image + Query

Hard without
tools?
 $p_0 \leq 0.5$

Yes

Keep query
cache baseline p_0

2 Domain-Specific Tool Use

Shared tools · Domain-conditioned evidence acquisition

Teacher-model agent loop

Reason

Tool call

Observe

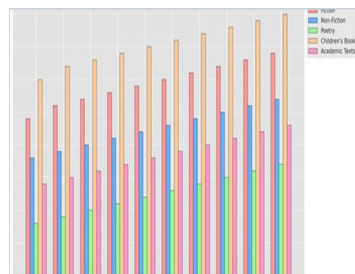
Image + Query

Tool Use

Observation + Answer

Chart Table

isolate
series
&
compute

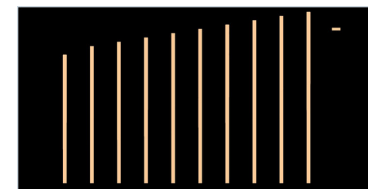


Query

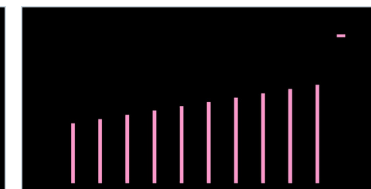
compute ratio:
Children's /
(Poetry +
Academic)

color_range

exec



Children's +50

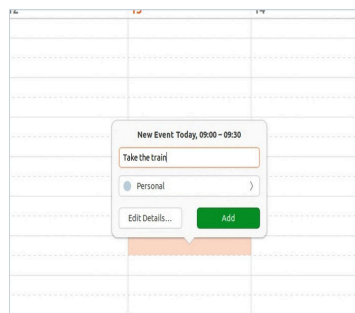


Poetry +45
Academic +45

Answer: $50 / (45 + 45) = 0.56$

GUI Visual Search

localize
then act

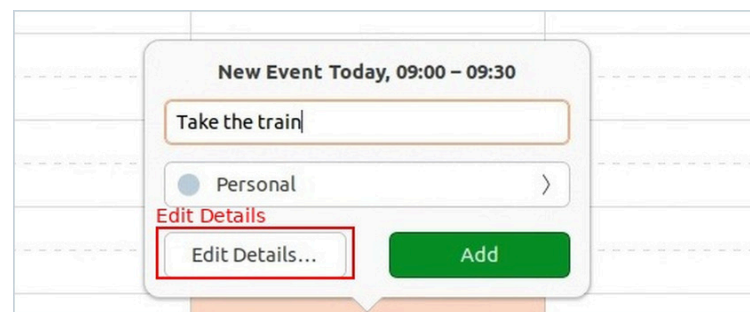


Query

Click
"Edit Details"

crop

draw_bbox



computer_use | left_click | [353, 453]

3 Supervision Verification

Keep traces that
pass both tests

Candidate trace

Both required

Outcome Validity

AND

Causal Utility



Instructive keep

OpenVisTool-42K